Fusing Computer Vision with Avian Morphology to Predict Flight Dynamics

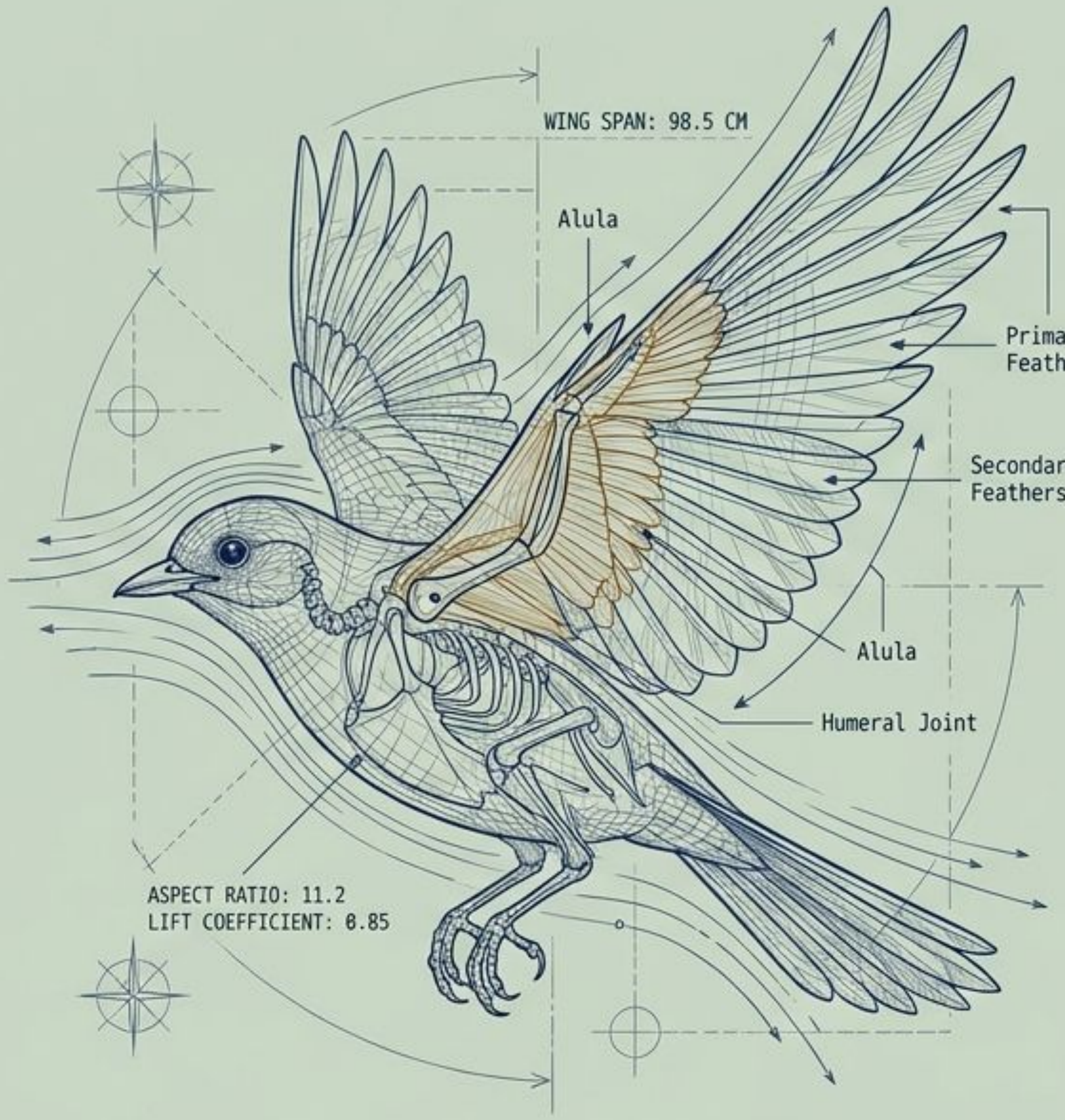
The AI Process

Motivation: Bird body structure and flight behavior are inextricably linked. Machine learning provides a powerful, quantifiable way to explore and explain this biological relationship.

Main Idea: A multimodal AI pipeline that ingests a user-uploaded bird image, predicts its taxonomic family, links it to tabular physical measurements, and computationally generates its likely flight style and wing model.

Results Briefly:
- Successfully adapted MobileNetV2 to classify five target bird families.
- Engineered morphological features (wing loading, aspect ratio) predicting speed with an R^2 of 0.956 (Ridge Regression).
- Achieved 99.6% accuracy classifying unsupervised K-means clusters into biologically accurate flight styles.

Biological Interpretation

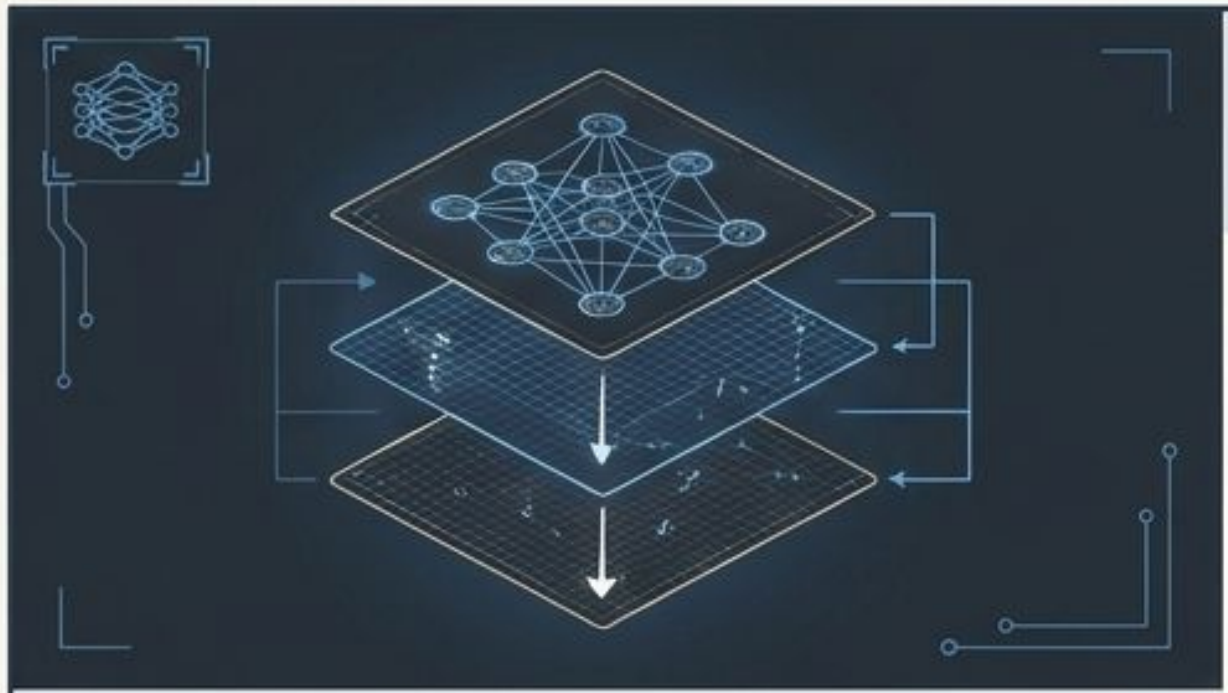


Literature review

This project uses transfer learning with MobileNetV2[1] for image-based bird-family classification, morphology-based feature engineering from AVONET[2] data, regression to estimate a speed index, and K-means clustering[3] to discover flight-style groups. The literature supports this combination because CNN transfer learning is effective for small image datasets, and avian morphology research[4] shows that wing loading, aspect ratio, and body mass are closely related to flight behavior.

1 - MobileNetV2: Inverted Residuals and Linear Bottlenecks - Mark Sandler - link
2 - AVONET: morphological, ecological and geographical data for all birds - Joseph A. Tobias - link
3 - Some methods for classification and analysis of multivariate observations - J. MacQueen - link
4 - Wing morphology, flight type and migration distance predict accumulated fuel load in birds - Orsolya Vincze - link

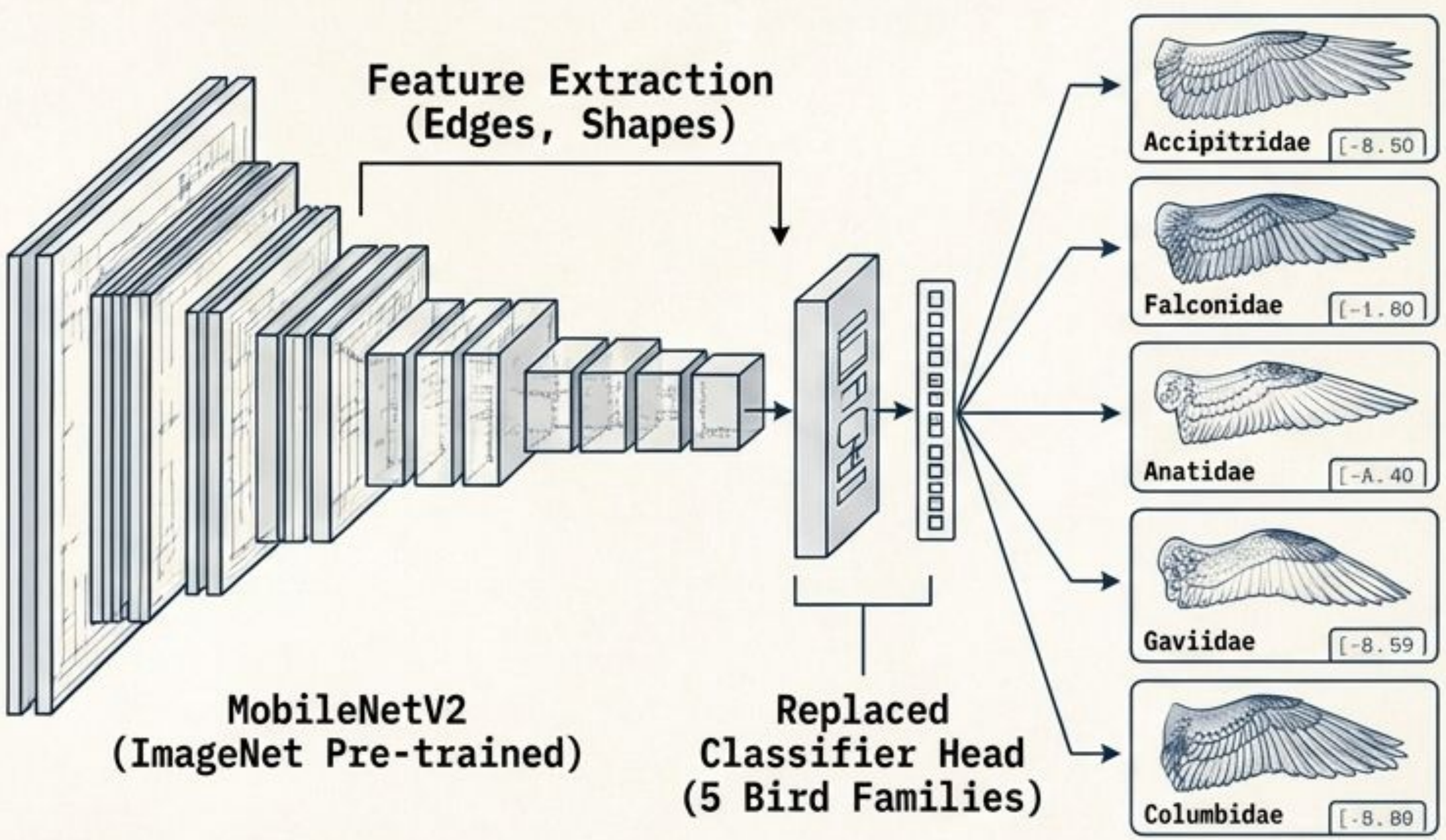
The Methodological Landscape: Core AI Concepts Applied - Transfer Learning



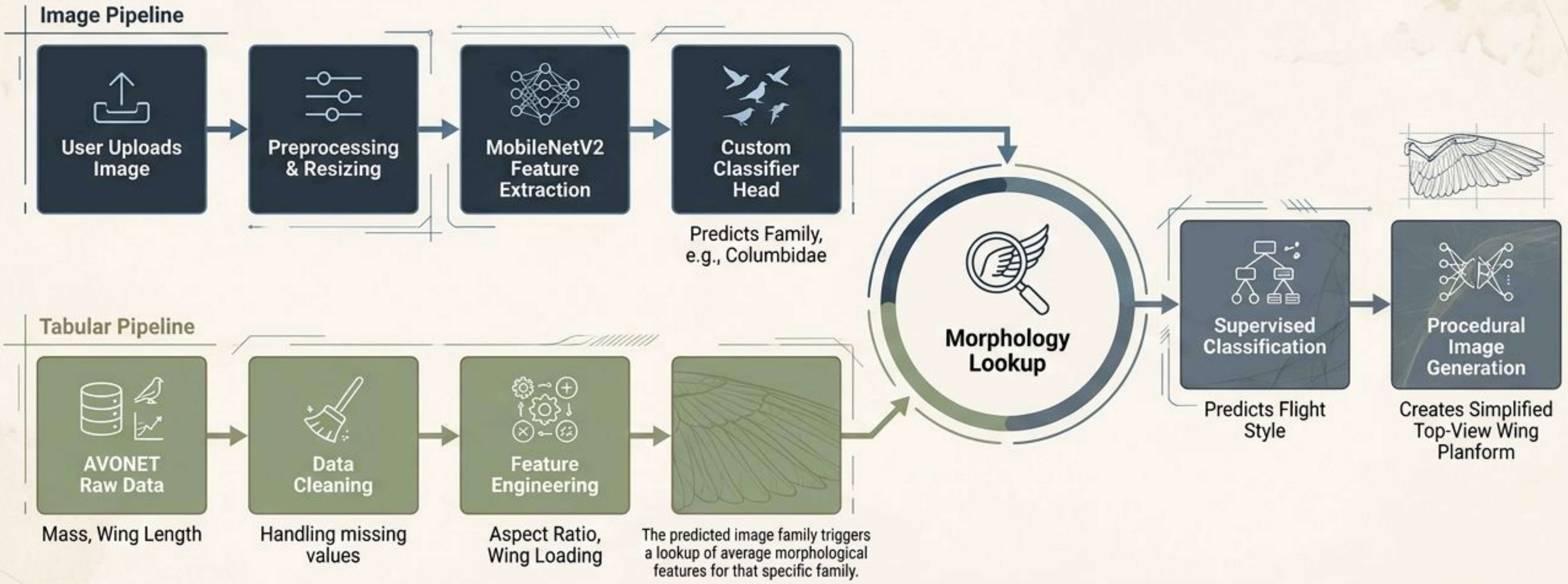
Method: Using models pre-trained on vast datasets for specific, smaller tasks.

Application: Utilizing MobileNetV2 (trained on ImageNet) to extract general visual patterns (edges, shapes) and replacing the classifier head to detect five specific bird families.

Transfer Learning

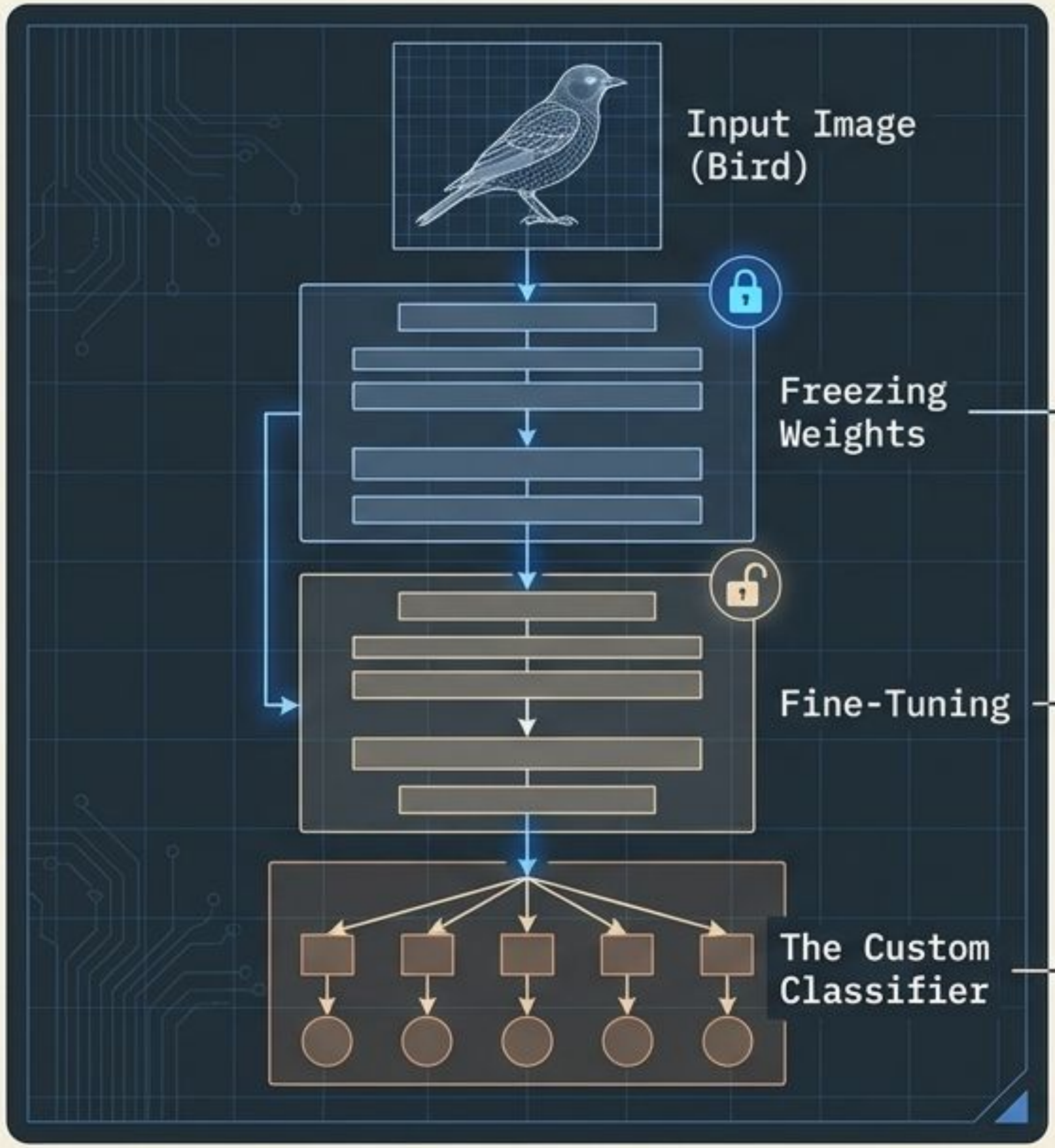


The Multimodal Architecture: Bridging Pixels and Physicality



Adapting MobileNetV2 for Avian Recognition via Transfer Learning

The Feature Extractor: The pre-trained body of the network. Early layers recognize general features (edges, curves); later layers recognize animal-like features (beaks, feathers).



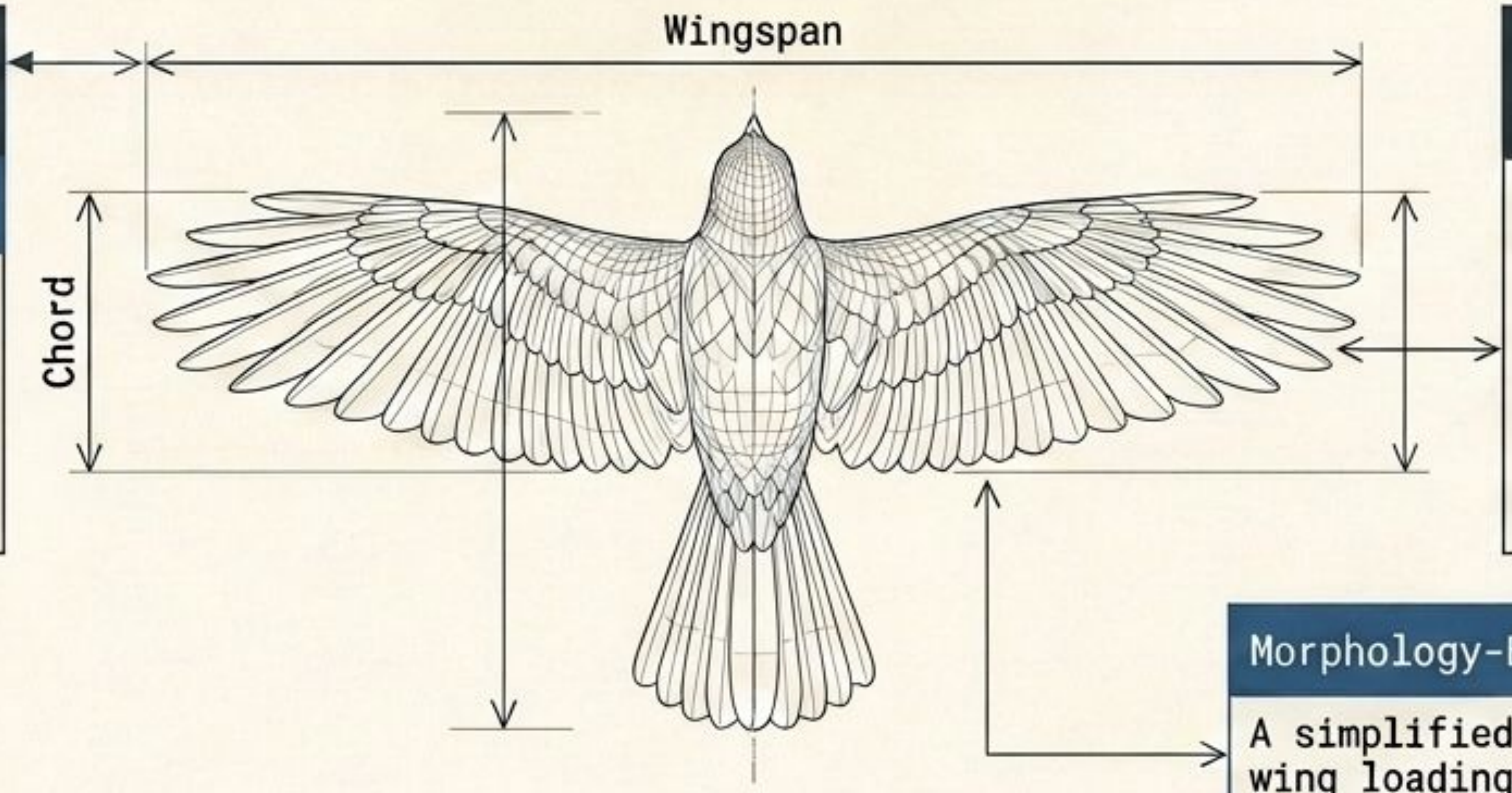
Initial training locks the pre-trained weights to prevent destroying useful general features, focusing solely on training the new final layers.

Unfreezing later layers with a small learning rate allows the model to specialize slightly to the new bird-image dataset without catastrophic forgetting.

The original ImageNet head is stripped and replaced. The new head outputs predictions for five target families: Accipitridae, Anatidae, Columbidae, Picidae, Strigidae.

Feature Engineering: Translating Raw Measurements into Flight Dynamics

Anatomy of a Wing



Aspect Ratio (Wingspan^2 / Wing Area):

Wing Area

Indicates wing shape.

Biological meaning: Long, narrow wings support efficient gliding/soaring.

Wing Loading (Weight / Wing Area):

Relationship between body mass and the lift surface.

Biological meaning: High loading requires faster, more power-demanding flight.

Morphology-Based Speed Index:

A simplified estimated score combining wing loading and Hand-Wing Index (HWI) to create a relative flight-performance target.

Data Transformation
Raw Measurements
Accipiter albogularis | Mass: 248.8g | Wing Length: 235.2mm

Validating Biological Relevance through Regression

Can our engineered morphology features reliably predict the estimated speed index?
Yes. The strong performance of simple linear models suggests the engineered features captured highly structured, meaningful biological relationships.

Bronze/Lowest

Gradient Boosting Regressor

$R^2 \approx 0.845$

Moderate performance.

Silver/Medium

Random Forest Regressor

$R^2 \approx 0.881$

Strong non-linear performance.

Gold/Highest

Ridge Regression (Optimal)

$R^2 \approx 0.956$

Mean Absolute Error ≈ 2.59

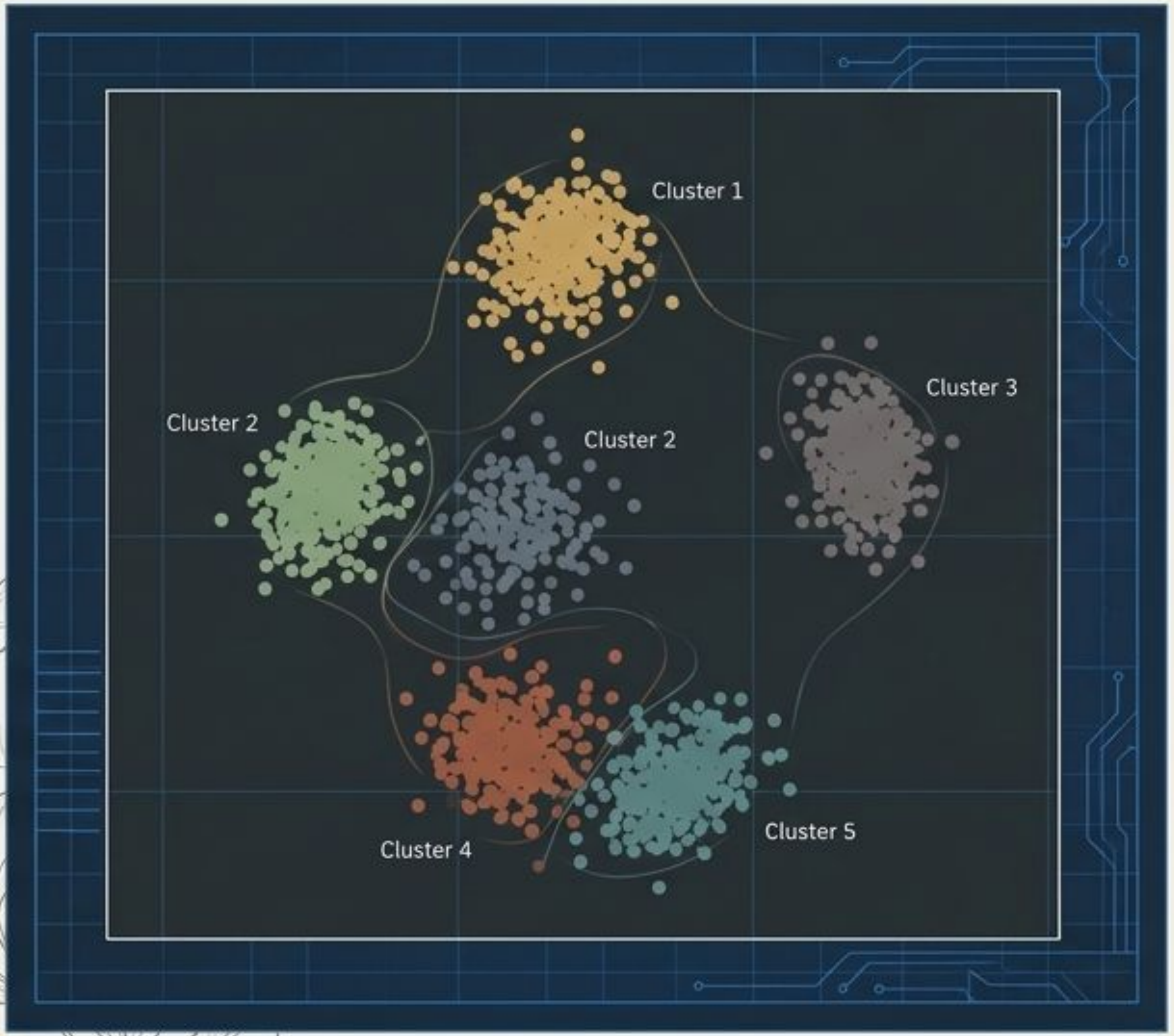
Winner

→ Takeaway: The linear simplicity of Ridge Regression outperforming complex ensembles indicates a direct, powerful correlation between physical traits (wing loading) and flight capability.

Gradient Boosting Regressor | Random Forest Regressor | Ridge Regression (Optimal)

Unsupervised Discovery: Extracting Flight Styles without Labels

Because real-world “flight style” labels didn’t exist in the AVONET dataset, K-Means clustering was deployed to group birds inherently based on feature similarity.



Cluster Mapping Table

Cluster	Biological Interpretation
Cluster 0 & 3 (Soaring / Gliding):	Maps strongly to Accipitridae (hawks/eagles) and Strigidae (owls). Broad wings adapted for riding air currents or slow, silent flight. 
Cluster 1 & 4 (Agile Maneuvering):	Maps to Picidae (woodpeckers) and Columbidae (pigeons/doves). Adapted for short, controlled paths through trees. 
Cluster 2 (Fast Direct Flyers):	Maps to Anatidae (ducks/geese). Driven by strong, continuous wingbeats. 

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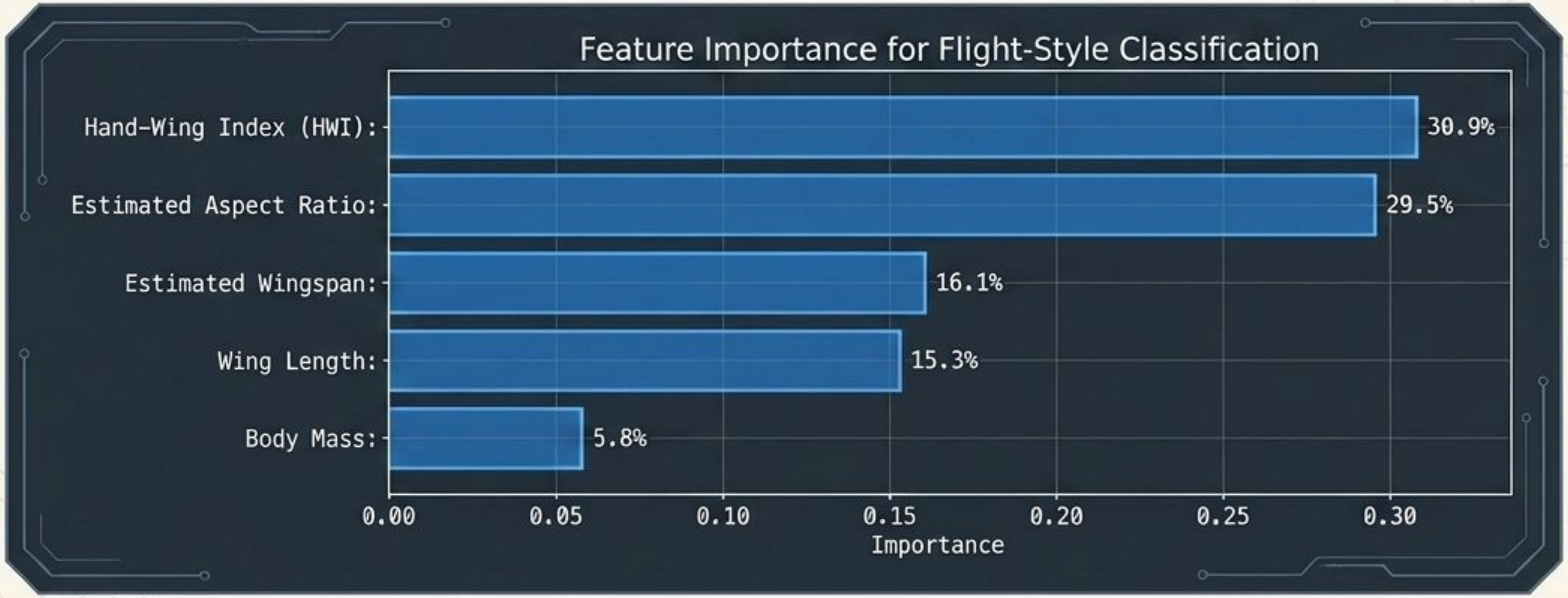
Supervised Classification & Model Interpretability

Once K-Means defined the five flight-style groups, a Random Forest Classifier was trained to predict these groups for any new bird. The pipeline achieves a 99.6% classification accuracy on the test set.



The AI learned that **wing shape (HWI, Aspect Ratio)** is vastly **more determinative** of flight style than **simple size** or **weight**.

Explainable AI: Translating Mathematics into English

A true AI pipeline must avoid the 'Black Box' problem. The system evaluates a specific bird's features against global medians to generate human-readable reasoning for its prediction.

The Code

```
def explain_bird(row, global_medians):
    explanations = []

    if row["aspect_ratio_est"] >
    global_medians["aspect_ratio_est"]:
        explanations.append("higher-than-average aspect ratio,
        suggesting longer and narrower wings.")
    else:
        explanations.append("lower-than-average aspect ratio,
        suggesting shorter or wider wings.")

    if row["wing_loading_N_m2"] >
    global_medians["wing_loading_N_m2"]:
        explanations.append("higher-than-average wing loading,
        which often indicates faster or more power-demanding
        flight.")
    else:
        explanations.append("lower-than-average wing loading,
        which can support slower, more efficient flight.")

    # ... (further conditional logic)
```

The Output

Example Case: *Accipiter albogularis*

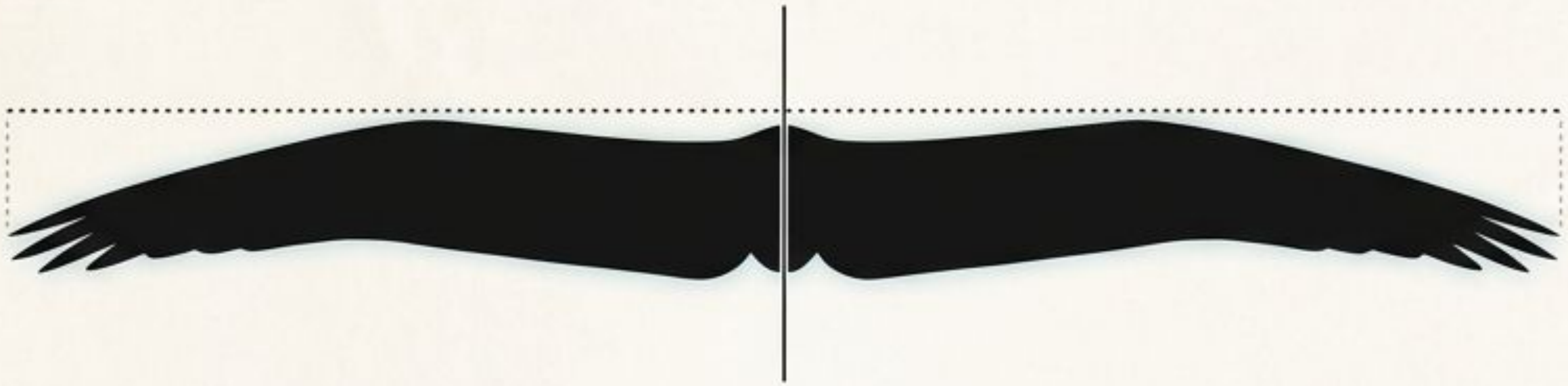
Predicted Flight Style: Soaring / gliding flyers

System Explanation:

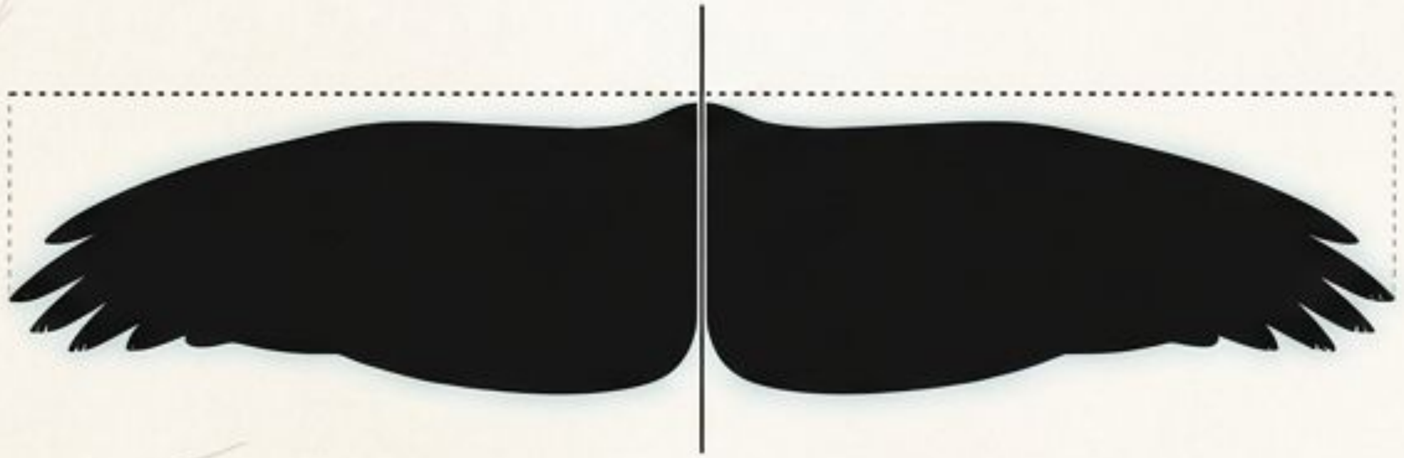
- Higher-than-average aspect ratio, suggesting longer and narrower wings.
- Higher-than-average wing loading, indicating faster or more power-demanding flight.
- Higher morphology-based speed index.
- Relatively high hand-wing index, associated with efficient long-distance flight.

Generative Visualization: Procedural Wing Modeling

The program features an image generator that translates the AI's continuous morphology values into a top-view, spread-wing planform. Even birds within the same flight cluster produce unique diagrams based on minor feature variations.



AR=15.21 | WL=145.5 | HWI=45.8



AR=8.94 | WL=102.1 | HWI=28.3

The Generative Ruleset:

Larger Estimated Wingspan
→ Longer displayed wings

Higher Aspect Ratio
→ Narrower effective planform

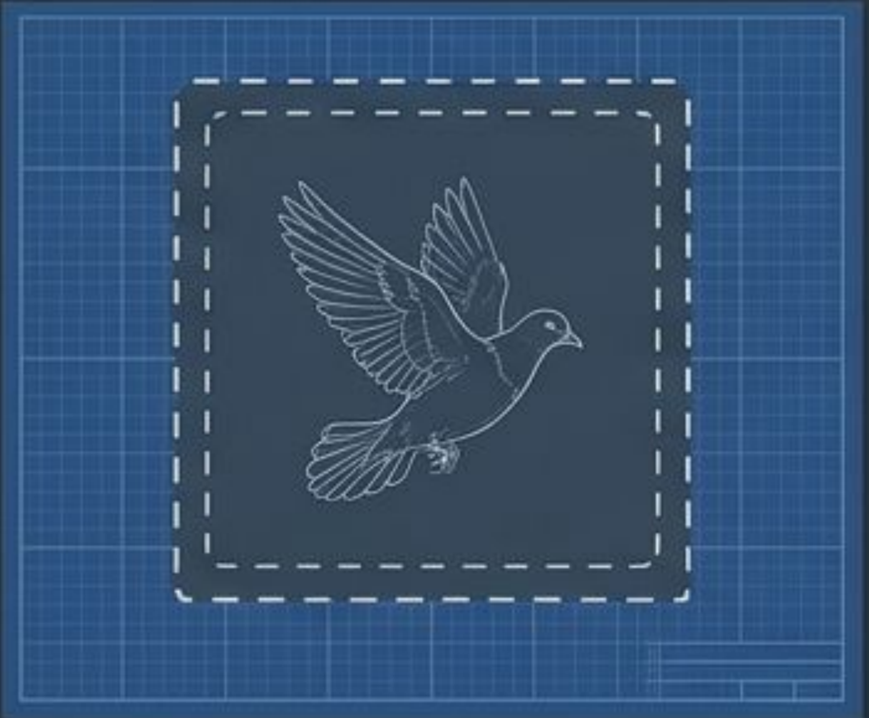
Higher Wing Loading
→ Thicker wing root

Higher HWI / Speed Index
→ Sharper, more swept wingtip

Lower HWI
→ Broader tips, visible feather notches

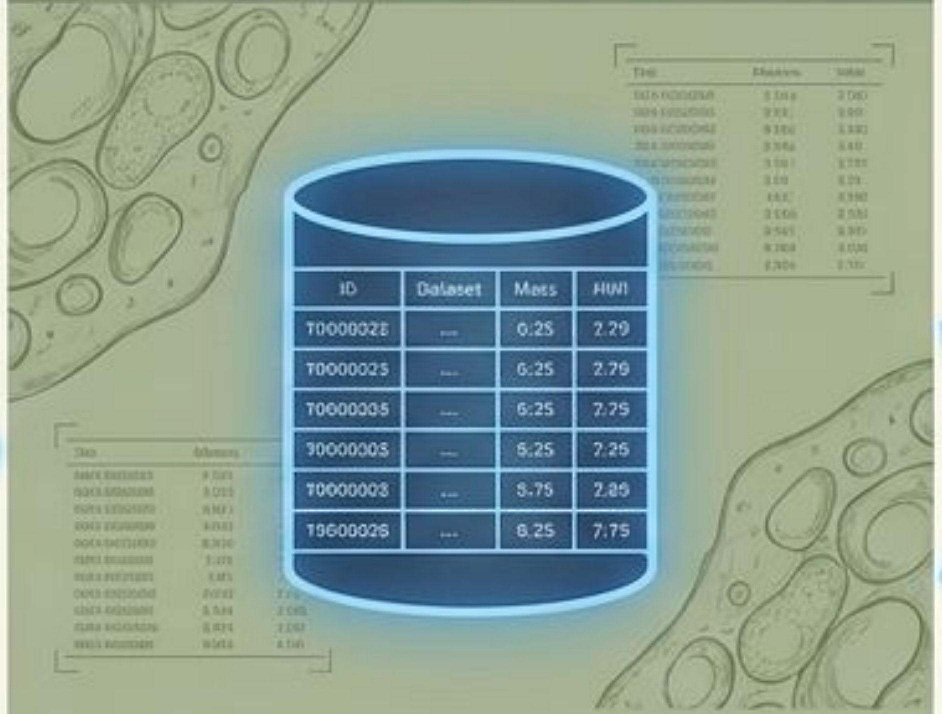
The Multimodal Pipeline in Action: A Complete Example

Image Recognition



User uploads an image. Computer Vision model predicts family: Columbidae.

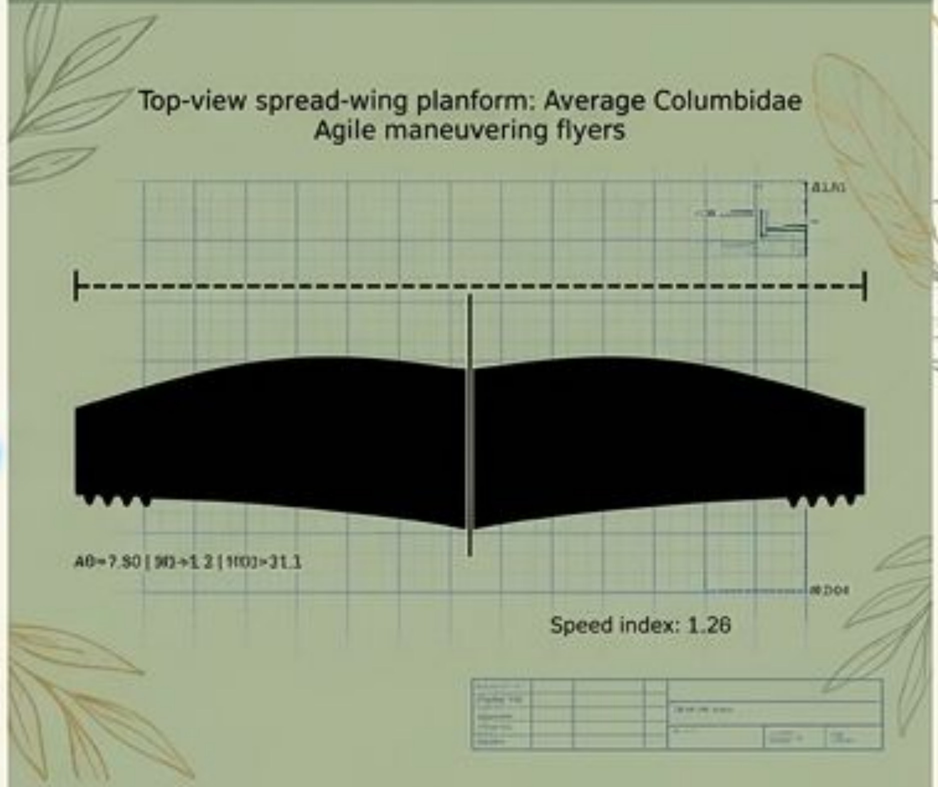
Morphology Integration



System queries AVONET dataset (353 rows used) to build an average Columbidae morphological profile (Mass: 0.25kg, Aspect Ratio: 7.79).

Random Forest predicts flight style: Agile maneuvering flyers.

Generative Output



The system renders the top-view spread-wing planform representing the average structural constraints of the predicted family.

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Future Explainability: Integrating Grad-CAM

While the tabular morphology data utilizes XAI techniques, the current Convolutional Neural Network (MobileNetV2) remains a black box.

Standard Image Classification Output

Prediction: Accipitridae

Grad-CAM Heatmap Overlay

Explanation: Focus on Head/Beak

The Grad-CAM Upgrade:

- Gradient-weighted Class Activation Mapping (Grad-CAM) highlights the specific pixel regions that influenced the CNN's prediction.
- Rather than just predicting 'Accipitridae,' the model will visually demonstrate whether it focused on the bird's beak, tail shape, or wing curvature to make its decision, serving as a sophisticated ML debugging tool.

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Critical Evaluation: Limitations & AI Ethics

Data Sparsity & Bias

The morphology dataset contained missing values, requiring aggressive filtering.

Skewed or sparse training data is the primary driver of biased predictions in ML systems.

Image classification quality degrades drastically with blurry or poorly angled photos.

Feature Limitations

The morphology-based “speed index” is an engineered mathematical proxy, not a real-world measured flight speed.

Future improvements require integrating real GPS tracking or observed migration data to align the model with reality.

Foundational Model Scale

Transfer learning relies on massive foundational models (like MobileNet).

As discussed in lecture, these models carry significant environmental compute costs and inherit intrinsic biases from their original, massive, uncurated training datasets.

Synthesis: AI as a Biological Translator

Machine learning is not just a tool for optimization; it is a mechanism for discovery. By fusing computer vision with **tabular feature engineering**, this project successfully bridges the gap between identifying what an animal IS and mathematically explaining how it FUNCTIONS.

– Physical structure and behavioral dynamics are mathematically linked.
– Multimodal pipelines provide significantly richer insights than single-model approaches.
– Explainability (XAI) transforms statistical predictions into legible biological intelligence.

Link to code

https://github.com/later8/ai-hw-spring-2026-nn/blob/7b9cd0f8ce109233094add0661d598105c5ac359/MiniRP/bird_flight_multimodal_ai_notebook_classification.ipynb

Youtube demo

https://youtu.be/1pKHqOmS_rA